

Shakil Mosharrof

Microsoft Certified: [AI Engineer \(AI-102\)](#), [Data Scientist \(DP-100\)](#), [MLOPs Engineer \(AI-300\)](#), [Databricks Data Engineer \(DP-750\)](#), [Databricks Certified: Machine Learning Engineer Professional](#), [NVIDIA : AIO](#), [Google Certified : Professional Cloud Architect](#) | [Researcher](#)

[Portfolio](#)
[GitHub](#)
[LinkedIn](#)

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SKILLS ↗

Language	Python, R, C++, C#, Bash, JavaScript, TypeScript, SQL
Backend	Flask, FastAPI, Redis, RQ, Celery, Django, Pydantic, Alembic, SQLAlchemy, Kafka
Database	MySQL, PostgreSQL, MongoDB, SQLite
Machine Learning	ML Metrics & evaluation, Supervised & Unsupervised learning, scikit-learn, XGboost, LightGBM, LLMs, PyTorch, NLP, TensorFlow, OpenCV, Yolo, GANs, CNN, Transformers, Agentic & LLM Integration & Development, Langchain Ecosystem, RAG, HuggingFace, Fine-Tuning Datasets with LLM and traditional ML Models, Prompt Engineering, ETL pipeline, W&B, AirFlow, DVC, MCP, MLflow
DevOps, MLOPs	CI/CD Pipelines, k8s, Docker, Terraform, On-Prem and Cloud
Agentic Tools	Claude Code (Pro), Codex (Pro), Cursor (Pro)

PROJECTS | CERTIFICATIONS ↗

MIST Mongol Barota (An autonomous rover) 2021 - 2023
Mars Society's University Rover Challenge (URC) USA
Anatolian Rover Challenge (ARC) UKET, Turkey
Tech Stack [ROS](#), [Robotics](#), [ML](#), [Autonomy](#), [Python](#), [Flask](#), [R&D](#)

Retrieval Augmented Generation Using Gemini Pro LLM and FastAPI | [Github](#)
Tech Stack [FastAPI](#), [Gemini](#), [LLM](#), [RAG](#), [API development](#)

Multi-Agent Data Analyst | [Github](#)
Tech Stack [LangGraph](#), [LLM](#), [FastAPI](#), [SQLAlchemy](#), [VectorDB](#)

FastAPI LLM RAG Backend Application on Medical Dataset (CSV) | [Github](#)
Tech Stack [LangGraph](#), [LLM](#), [FastAPI](#), [SQLAlchemy](#), [VectorDB](#)

Predicting Bike-sharing Demand Using Machine Learning with AutoGluon | [Github](#)
Tech Stack [MLops](#), [Sagemaker](#), [AutoGluon](#), [Torch](#), [MaxNet](#)

Automatic Literature Review generation with RAG Based LLMs | [Github](#)
Tech Stack [LLMs](#), [RAG](#), [OpenAI API](#), [HuggingFace Models](#)

PyTorch-Create Your Own Classifier | [Github](#) | [Certificate](#)
Tech Stack [Python](#), [CNN](#), [Machine Learning](#), [AWS](#), [Udacity](#)

MNIST Neural Network Project using pyTorch | [Github](#)
Tech Stack [PyTorch](#), [Python](#), [CNN](#), [Machine Learning](#)

RESEARCH & PUBLICATIONS (7+) ↗ | SCHOLAR ↗

- DeepTriNet: A Tri-Level Attention Based DeepLabv3+ Architecture for Semantic Segmentation of Satellite Images. [\[Link\]](#) [got the best paper award]
- Performance Analysis of Various EfficientNet Based U-Net++ Architecture for Automatic Building Extraction from High Resolution Satellite Images. [\[Link\]](#)
- CycleGAN-Based Data Augmentation with CNN and Vision Transformers (ViT) Models for Improved Maize Leaf Disease Classification. [\[Link\]](#)
- Automatic Literature Review Generation : An Improved LLM-RAG Based Solution Embedded With Regex and T5. [Link](#) [Thesis] [ICISSE-2024]

EXPERIENCE ↗

Senior AI Engineer (Agentic AI & MLOPs) **April 2025 - Now**
GenerexAI FL, USA, Remote, Contract

- Built & maintained an agentic AI backend for US home remodeling customer support & real estate workflow automation, using a custom Asynchronous Multi-Agent Architecture (AMA) with FastAPI microservices to support multi-step reasoning, tool execution, async task orchestration, & end-to-end QA coverage across agent workflows with aws deployment and CI/CD.
- Designed & owned an event-driven billing & notification platform integrating Stripe & Orb Billing, with async webhook processing, database-level idempotency, exponential backoff retries, dead-letter queues, reconciliation pipelines, & multi-channel notifications across email, SMS, & in-app alerts..

Machine Learning Engineer **December 2024 - March 2025**
Culture Hint Remote (Freelance) London, UK

- Improving & implementing machine learning & computer vision solutions with back-end API designing with AWS EC2, CI/CD Pipeline, YOLOv8, Labelstudio

AI Engineer **July 2024 - December 2024**
Delineate Pro Inc. (YC'25) Remote (Contract) Massachusetts, USA

- Developed & delivered end to end LLM based RAG solutions for major pharmaceutical companies in the USA, enhancing their research workflows & decision-making processes.
- Implemented cloud-based Retrieval-Augmented Generation (RAG) agents to optimize data retrieval & processing, significantly improving research efficiency & accuracy for pharmaceutical applications.

Software Engineer (AI/ML) **March - August 2024**
Gigalogy Inc. Hybrid (BD), Tokyo, Japan

- Engineered scalable RESTful APIs using FastAPI, integrating SQLAlchemy, PyDantic, & Docker, while ensuring robust performance through Insomnia API testing.
- Designed & deployed a RAG LLM-based recommender system for a Japanese E-Commerce platform, significantly enhancing product recommendation accuracy & user engagement.

Software & Autonomous System Developer (Onsite) 2021 - 2023
Mars Society's University Rover Challenge (URC) USA
Anatolian Rover Challenge (ARC) UKET, Turkey

- Led the development of an autonomous rover system, securing 1st place at the University Rover Challenge (URC 2021) & 3rd place at the Anatolian Rover Challenge (ARC 2022), competing against teams from over 14 countries worldwide. [Link](#) , [Link](#).
- Achieved over \$50,000 in funding from government & non-government sources, contributing to the success of projects in URC 2023 & ARC 2023, where my team placed 2nd in Asia & 5th globally.

EDUCATION ↗

B.Sc in Computer Science and Engineering
Military Institute of Science and Technology
Higher Secondary School Certificate
Barishal Cadet College